

reachable cosine

0.63

held-out (realized)

0.45

held-out z

45

confidence

HIGH

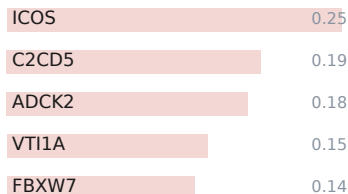
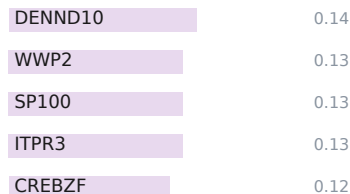
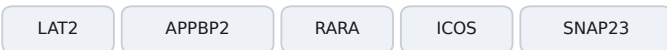
How the target splits by intervention direction



knockdown

activation

out of reach

KNOCKDOWN recipe (CRISPRi)**ACTIVATION recipe (CRISPRa)****NEXT SCREEN — optimal library (k=5)**

Druggable knockdown targets: 7 small-molecule · 1 clinical-stage

39% of the addressable shift needs INDUCTION (CRISPRa/ORF); a knockdown-only screen structurally cannot reach it. 1/15 knockdown targets have an approved/clinical drug for repurposing.

→ Th1 · Stim8hr

WEAKLY REACHABLE

reachable cosine

0.52

held-out (realized)

0.29

held-out z

13

confidence

MEDIUM

How the target splits by intervention direction



knockdown

activation

out of reach

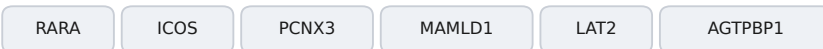
KNOCKDOWN recipe (CRISPRi)

AGTPBP1	0.17
RARA	0.15
HERC1	0.14
DPH1	0.14
MAMLD1	0.13

ACTIVATION recipe (CRISPRa)

ZBTB49	0.18
AP5S1	0.16
PEX16	0.13
APOBEC3F	0.13
TXNDC5	0.12

NEXT SCREEN — optimal library (k=6)



Druggable knockdown targets: 6 small-molecule · 2 clinical-stage

49% of the addressable shift needs INDUCTION (CRISPRa/ORF); a knockdown-only screen structurally cannot reach it. 2/15 knockdown targets have an approved/clinical drug for repurposing.

→ Th1 · Stim48hr

WEAKLY REACHABLE

reachable cosine

0.53

held-out (realized)

0.30

held-out z

22

confidence

MEDIUM

How the target splits by intervention direction



knockdown

activation

out of reach

KNOCKDOWN recipe (CRISPRi)

FBXO32	0.23
ICOS	0.17
KLHL6	0.17
RARA	0.17
ATG5	0.15

ACTIVATION recipe (CRISPRa)

GBP5	0.15
ADD1	0.14
SAMD8	0.12
MYH9	0.12
ZFP36	0.12

NEXT SCREEN — optimal library (k=4)

ATF7IP2

FBXO32

RARA

HPS1

Druggable knockdown targets: 4 small-molecule · 3 clinical-stage

48% of the addressable shift needs INDUCTION (CRISPRa/ORF); a knockdown-only screen structurally cannot reach it. 3/15 knockdown targets have an approved/clinical drug for repurposing.

reachable cosine

0.64

held-out (realized)

0.46

held-out z

21

confidence

HIGH

How the target splits by intervention direction



knockdown

activation

out of reach

KNOCKDOWN recipe (CRISPRi)

MAPK14	0.21
IRF1	0.18
WWP2	0.17
ILVBL	0.17
SIN3A	0.16

ACTIVATION recipe (CRISPRa)

ADCK2	0.14
ICOS	0.14
CCND3	0.13
ZBTB22	0.12
C2CD5	0.12

NEXT SCREEN — optimal library (k=4)

IRF1

RSBN1L

DENND4B

SFXN1

Druggable knockdown targets: 6 small-molecule · 2 clinical-stage

36% of the addressable shift needs INDUCTION (CRISPRa/ORF); a knockdown-only screen structurally cannot reach it. 2/15 knockdown targets have an approved/clinical drug for repurposing.

reachable cosine

0.54

held-out (realized)

0.30

held-out z

20

confidence

MEDIUM

How the target splits by intervention direction



knockdown

activation

out of reach

KNOCKDOWN recipe (CRISPRi)

IFNGR1	0.22
ZBTB49	0.17
DZIP1	0.14
GGA1	0.13
PACS1	0.13

ACTIVATION recipe (CRISPRa)

AMPD3	0.13
SEPTIN2	0.13
BST2	0.12
MAGEH1	0.12
MAL	0.12

NEXT SCREEN — optimal library (k=5)

DENND4B

JAK2

IL7R

IRF9

MAPK14

Druggable knockdown targets: 6 small-molecule · 2 clinical-stage

46% of the addressable shift needs INDUCTION (CRISPRa/ORF); a knockdown-only screen structurally cannot reach it. 2/15 knockdown targets have an approved/clinical drug for repurposing.

reachable cosine

0.53

held-out (realized)

0.26

held-out z

13

confidence

MEDIUM

How the target splits by intervention direction



knockdown

activation

out of reach

KNOCKDOWN recipe (CRISPRi)

IRF1	0.23
TMSB15B	0.18
GFI1	0.17
GBP5	0.16
DNTTIP1	0.16

ACTIVATION recipe (CRISPRa)

FBXO32	0.18
AMPD3	0.12
ATG5	0.12
ICOS	0.12
YWHAQ	0.12

NEXT SCREEN — optimal library (k=4)

RSBN1L

TMSB15B

IRF1

TAFAZZIN

Druggable knockdown targets: 4 small-molecule · 2 clinical-stage

48% of the addressable shift needs INDUCTION (CRISPRa/ORF); a knockdown-only screen structurally cannot reach it. 2/15 knockdown targets have an approved/clinical drug for repurposing.

→ younger · Rest

PARTIALLY REACHABLE

reachable cosine

0.63

held-out (realized)

0.40

held-out z

9

confidence

LOW

How the target splits by intervention direction



knockdown

activation

out of reach

KNOCKDOWN recipe (CRISPRi)

COPS7B	0.09
CCND3	0.09
SEPTIN7	0.09
CLINT1	0.09
HSPA1L	0.09

ACTIVATION recipe (CRISPRa)

TYW5	0.08
AGAP3	0.08
ZNF544	0.08
NAA80	0.08
PDE4B	0.07

NEXT SCREEN — optimal library (k=4)

FAM98A

DEPDC5

NFAT5

COIL

Druggable knockdown targets: 6 small-molecule · 0 clinical-stage

36% of the addressable shift needs INDUCTION (CRISPRa/ORF); a knockdown-only screen structurally cannot reach it. 0/15 knockdown targets have an approved/clinical drug for repurposing.

→ younger · Stim8hr

WEAKLY REACHABLE

reachable cosine

0.59

held-out (realized)

0.35

held-out z

5

confidence

LOW

How the target splits by intervention direction



knockdown

activation

out of reach

KNOCKDOWN recipe (CRISPRi)

KLF3	0.11
VANGL1	0.09
CAMK1	0.08
JMJD4	0.07
VCPKMT	0.07

ACTIVATION recipe (CRISPRa)

C3orf38	0.08
SNRNP40	0.06
ABRAXAS1	0.06
AMER1	0.06
CD320	0.06

NEXT SCREEN — optimal library (k=4)

TRIT1

HVCN1

PAN3

ALKBH1

Druggable knockdown targets: 5 small-molecule · 0 clinical-stage

40% of the addressable shift needs INDUCTION (CRISPRa/ORF); a knockdown-only screen structurally cannot reach it. 0/15 knockdown targets have an approved/clinical drug for repurposing.

→ younger · Stim48hr

WEAKLY REACHABLE

reachable cosine

0.60

held-out (realized)

0.36

held-out z

6

confidence

LOW

How the target splits by intervention direction

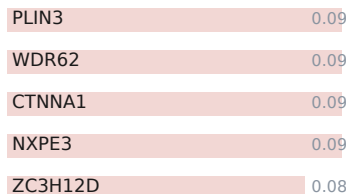


knockdown

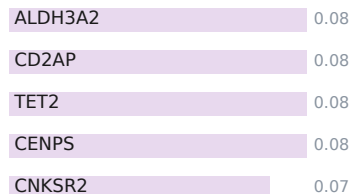
activation

out of reach

KNOCKDOWN recipe (CRISPRi)



ACTIVATION recipe (CRISPRa)



NEXT SCREEN — optimal library (k=4)

SLC9A1

ATXN2

MFSD10

PLIN3

Druggable knockdown targets: 3 small-molecule · 0 clinical-stage

39% of the addressable shift needs INDUCTION (CRISPRa/ORF); a knockdown-only screen structurally cannot reach it. 0/15 knockdown targets have an approved/clinical drug for repurposing.

→ older · Rest

WEAKLY REACHABLE

reachable cosine

0.60

held-out (realized)

0.39

held-out z

14

confidence

MEDIUM

How the target splits by intervention direction



knockdown

activation

out of reach

KNOCKDOWN recipe (CRISPRi)

KDM6A	0.10
AGAP3	0.09
MAP3K4	0.09
PRPF38B	0.08
TYW5	0.08

ACTIVATION recipe (CRISPRa)

TMEM265	0.07
H2AC11	0.07
HSPBP1	0.07
SEPTIN7	0.07
COPS7B	0.07

NEXT SCREEN — optimal library (k=4)

C1D

SMG1

SUPT7L

TARS2

Druggable knockdown targets: 9 small-molecule · 2 clinical-stage

41% of the addressable shift needs INDUCTION (CRISPRa/ORF); a knockdown-only screen structurally cannot reach it. 2/15 knockdown targets have an approved/clinical drug for repurposing.

→ older · Stim8hr

WEAKLY REACHABLE

reachable cosine

0.59

held-out (realized)

0.36

held-out z

12

confidence

MEDIUM

How the target splits by intervention direction

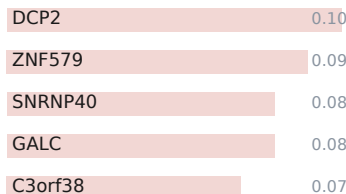


knockdown

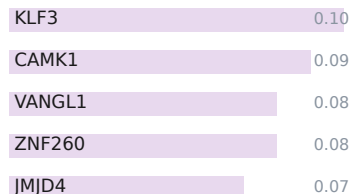
activation

out of reach

KNOCKDOWN recipe (CRISPRi)



ACTIVATION recipe (CRISPRa)



NEXT SCREEN — optimal library (k=4)



Druggable knockdown targets: 5 small-molecule · 1 clinical-stage

39% of the addressable shift needs INDUCTION (CRISPRa/ORF); a knockdown-only screen structurally cannot reach it. 1/15 knockdown targets have an approved/clinical drug for repurposing.

→ older · Stim48hr

WEAKLY REACHABLE

reachable cosine

0.57

held-out (realized)

0.33

held-out z

7

confidence

LOW

How the target splits by intervention direction



knockdown

activation

out of reach

KNOCKDOWN recipe (CRISPRi)

TRAF4	0.08
RASGRP1	0.08
VCF2	0.08
CNKSR2	0.08
RMI1	0.08

ACTIVATION recipe (CRISPRa)

TBC1D10A	0.07
NFATC2IP	0.07
COX7A2L	0.07
CTNNA1	0.07
FZD3	0.06

NEXT SCREEN — optimal library (k=5)

ZAP70

SMG1

CROT

WDFY1

CD5

Druggable knockdown targets: 7 small-molecule · 2 clinical-stage

44% of the addressable shift needs INDUCTION (CRISPRa/ORF); a knockdown-only screen structurally cannot reach it. 2/15 knockdown targets have an approved/clinical drug for repurposing.